AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS 2021

Chemistry

Total Time: 2 hours 10 minutes

Total Marks: 65 (50-Theory & 15-Alternate to Practical)

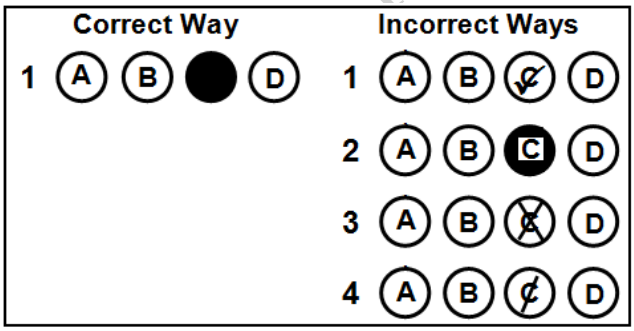
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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 65 only.
4. Question Distribution:

Theory	Alternate to Practical (ATP)
50 MCQs	15 MCQs

5. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
7. DO NOT write anything in the answer grid. The computer only records what is in the circles.
8. The marks obtained on the 50 MCQs will be equated to the total marks of 85 for the theory examination results.
9. You may use a scientific calculator if you wish.

THEORY (Questions 1-50)

1. If the molecular mass of an organic compound is 60 amu and its empirical formula is CH_2O, then its molecular formula will be

   A. CH_2O
   B. C_2HO_2
   C. $C_2H_4O_2$
   D. $C_2H_2O_2$

2. The number of bromide ions in 3.5 moles of $MgBr_2$ is

   (**Note**: Atomic mass of Br = 79.904 ≈ 80 amu)

   A. 1.204×10^{24} ions.
   B. 2.107×10^{24} ions.
   C. 4.214×10^{24} ions.
   D. 96.32×10^{24} ions.

3. Which of the following does NOT signify quantitative analysis?

   A. Predicting the expected amount of a product
   B. Calculating the actual yield of any reaction
   C. Determining the presence of lead in paint
   D. Measuring nutrient levels in blood

4. If '**X**' g of C and '2**X**' g of Mg contains the same number of atoms, i.e., 6.02×10^{23}, then **X** is equal to

   A. 6 g.
   B. 12 g.
   C. 24 g.
   D. 48 g.

5. How many grams of nitrogen is approximately present in every 100 g of ammonium sulphate $(NH_4)_2SO_4$?

   (**Note**: ${}^{1}_{1}H$, ${}^{14}_{7}N$, ${}^{16}_{8}O$, ${}^{32}_{16}S$)

   A. 132 g
   B. 114 g
   C. 28 g
   D. 21 g

6. If 100 molecules of acetic acid are ionised into acetate and hydronium ions out of 1000 molecules, then the number of ions produced by the dissociation of 100 moles of acetic acid will be

   A. 6.02×10^{23}
   B. 6.02×10^{24}
   C. 6.02×10^{25}
   D. 6.02×10^{26}

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7. The magnetic quantum number value(s) for 3d orbitals is/ are

   A. 0
   B. -1, 0, +1
   C. -2, -1, 0, +1, +2
   D. -3, -2, -1, 0, +1, +2, +3

8. In a discharge tube, the positive rays are produced when high speed electrons strike the

   A. anode with large atomic mass.
   B. glass wall opposite to the cathode.
   C. molecules of a gas enclosed in the tube.
   D. radioactive polonium source directed at beryllium target.

9. "It is impossible to determine the exact position and the momentum of a moving electron around the nucleus of an atom, simultaneously."

   This statement represents

   A. Aufbau principle.
   B. Bohr's atomic model.
   C. Pauli's exclusion principle.
   D. Heisenberg's uncertainty principle.

10. Henry Moseley did a systematic study of X-rays in the year 1913. According to his study, X-rays of shorter wavelength can be generated by

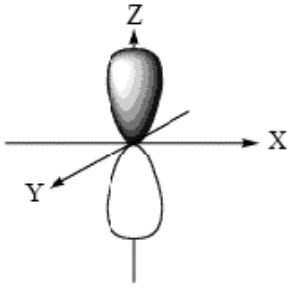
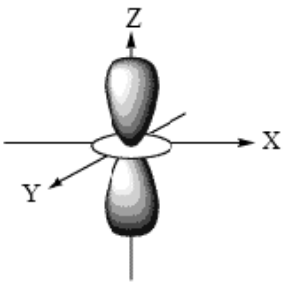
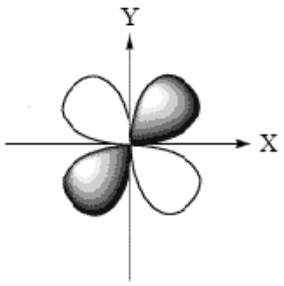
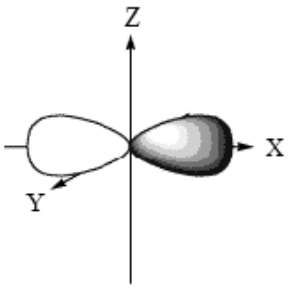
    A. using an anode of high atomic number.
    B. using a cathode of high atomic number.
    C. decreasing the voltage of X-ray generator.
    D. increasing the pressure of X-ray generator.

11. The de-Broglie wavelength of particle CANNOT be calculated by

    A. $\lambda = \frac{c}{\nu}$
    B. $\lambda = \frac{h}{m\nu}$
    C. $\lambda = \frac{h}{\sqrt{2 \times K.E \times m}}$
    D. $\lambda = \frac{h}{\sqrt{2qvm}}$

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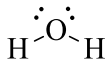
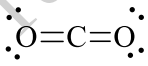
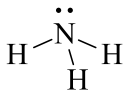
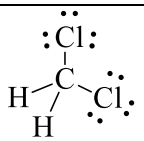
12. The CORRECT representation of $3d_z^2$ atomic orbital is

	
A	B
	
C	D

13. Which of the following orbitals of oxygen hybridise to form four sp^3 hybrid orbitals in water molecule?

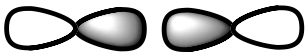
A. One 2s and three 2p
B. One 1s and three 2p
C. One 3s and three 3p
D. One 1s and three 3p

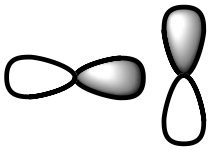
14. The molecule that shows zero net dipole moment is

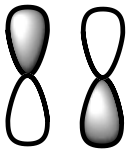
			
A	B	C	D

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15. Given are the pairs of orbitals showing overlaps.

I. 

II. 

III. 

The overlap(s) that will result in the σ3p molecular orbital is/ are

A. I only.
B. III only.
C. I and II.
D. II and III.

16. All of the following reasons are true about the directional nature of covalent bond EXCEPT that the

A. shared electrons are strongly pulled by the electronegative atom.
B. bonded atoms prefer specific orientation in space.
C. number of anions surrounding a cation is limited.
D. possible orbital overlaps are few in number.

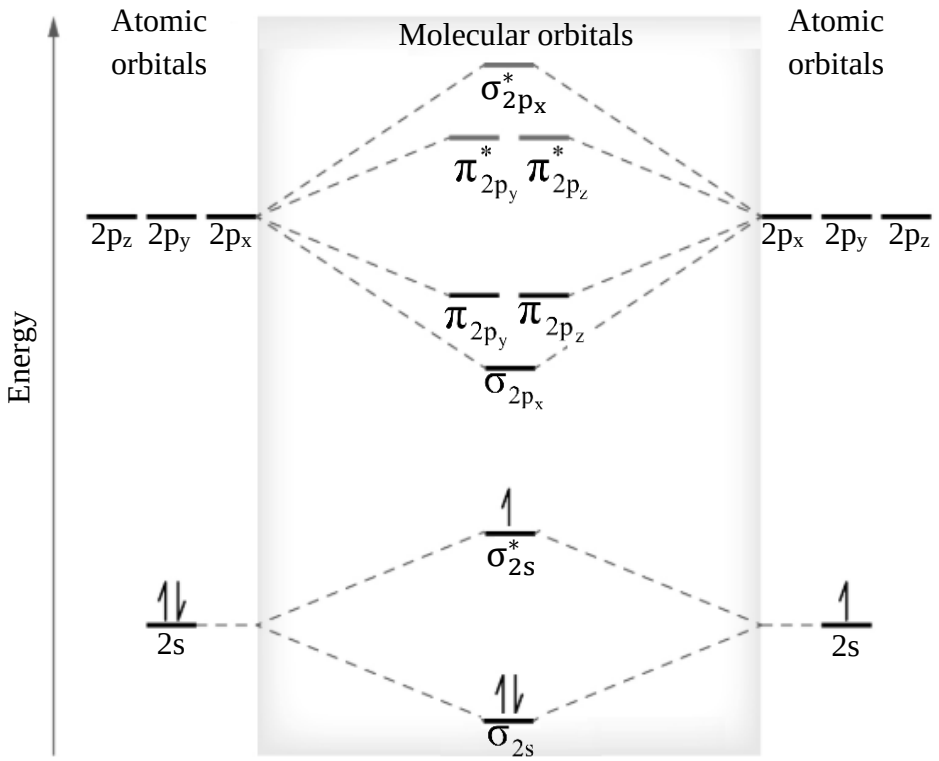
17. The molecular geometry of ammonia according to valence shell electron pair repulsion (VSPER) theory is

(**Note**: $^{14}_{7}$N and $^{1}_{1}$H)

A. angular.
B. tetrahedral.
C. trigonal planar.
D. trigonal pyramidal.

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18. Given is the molecular orbital diagram for Be_2^+ ion, showing the molecular orbitals of the valence shell only.



According to the given diagram, the bond order of Be_2^+ is

A. 0.5
B. 1.0
C. 1.5
D. 2.0

19. Water boils at low temperature when there is low

A. volume.
B. polarisation.
C. vapour pressure.
D. external pressure.

20. At boiling point of a pure liquid, until the entire liquid is converted into gas, the temperature of the system will

A. increase.
B. decrease.
C. remain constant.
D. initially decrease and then increase.

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21. The option that is CORRECT about the number of hydrogen bonds and the amount of energy required by a highly volatile liquid is

	Number of Hydrogen Bonds	Amount of Energy
A	Less	Minimum
B	Less	Maximum
C	More	Minimum
D	More	Maximum

22. In electronic display devices, the layer which is sandwiched between two sheets of polarised glass is composed of

A. gas.
B. plasma.
C. liquid crystals.
D. crystalline solids.

23. Anisotropy is the property of crystalline solid in which

A. few crystals show repetition of angles on 360° rotation along its axis.
B. some crystals show variations in physical properties with direction.
C. two different substances exist in the same crystalline form.
D. a compound exists in more than one crystalline form.

24. In the given table, the type of solid that CORRECTLY matches with its property is

	Type of Solid	Property
A	ionic	good conductors of heat and electricity
B	covalent	malleable and ductile
C	metallic	low melting point and high volatility
D	molecular	soft and easily compressible

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25. Given table shows two different phenomena, **I** and **II**, in which crystal lattices exist.

I		**II**	
Crystal Lattices of an Ionic Solid		**Crystal Lattices of an Element**	
Rhombohedral	Orthorhombic	Cubic	Hexagonal

The option that CORRECTLY identifies these phenomena is

	I	**II**
A	allotropy	polymorphism
B	polymorphism	isomorphism
C	isomorphism	allotropy
D	polymorphism	allotropy

26. The given crystal lattice is identified as



A. triclinic.
B. tetragonal.
C. hexagonal.
D. monoclinic.

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27. The dissociation of a weak acid, hydrogen sulphide (H_2S), in water can be suppressed by the addition of

A. sodium chloride (NaCl).
B. hydrogen chloride (HCl).
C. potassium chloride (KCl).
D. ammonium chloride (NH_4Cl).

28. The condition that only affects the rate of reaction without disturbing the direction of a reversible reaction is

A. catalyst.
B. pressure.
C. temperature.
D. concentration.

29. The equilibrium constant value for the given reaction is 1×10^{-13}.

$2HF_{(g)} \rightleftharpoons H_{2(g)} + F_{2(g)}$

Which of the following options is CORRECT for the given reaction?

	Value of K_c	**Stability of Reactant**
A	Very large	Unstable
B	Very small	Stable
C	Very large	Stable
D	Very small	Unstable

30. **X** is the property of a substance that shows its dissolution in a solvent and dependency on temperature, pressure and pH.

Based on the given statement, the property **X** is identified as

A. equilibrium constant.
B. common ion effect.
C. solubility product.
D. solubility.

31. The unit of equilibrium constant (K_c) for the given reaction is

$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

A. mol/dm^3.
B. dm^3/mol.
C. mol^6/dm^2.
D. mol^2/dm^6.

32. Consider the given esterification reaction between acetic acid and ethanol, for which the value of K_c at 25°C is 4. To check the progress of this reaction, a sample from the reaction mixture is drawn and the value of K_c is calculated as 2.5.

$$CH_3COOH + C_2H_5OH \overset{H^+}{\rightleftharpoons} CH_3COOC_2H_5 + H_2O$$

The calculated value of K_c depicts that the reaction

A. is completed.
B. is at equilibrium.
C. will proceed in the forward direction to attain equilibrium.
D. will proceed in the backward direction to attain equilibrium.

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33. In the given reaction, the reactant H_2O can be classified as a/ an

$$Cr^{3+}_{(aq)} + 6H_2O_{(l)} \rightleftharpoons Cr(H_2O)^{3+}_{6(aq)}$$

A. Lewis acid.
B. Lewis base.
C. neutral species.
D. amphoteric species.

34. When methanol reacts with water, it acts as a Brønsted-Lowry acid with the equilibrium position lying very far to the left.

$$CH_3OH + H_2O \rightleftharpoons CH_3O^- + H_3O^+$$

Which of the following options CORRECTLY depicts the strength of acid and its conjugate base in the given reaction?

	Acid (CH_3OH)	**Conjugate Base (CH_3O^-)**
A	Weaker	Weaker
B	Stronger	Weaker
C	Weaker	Stronger
D	Stronger	Stronger

35. The pH of a sample containing 0.0125 M solution of potassium hydroxide (KOH) is

A. 1.90
B. 3.61
C. 12.10
D. 13.98

36. Which of the following statements is TRUE about the buffer solutions?

A. Their pH alters on dilution.
B. They have a specific constant value of pH.
C. They show a change in pH on addition of an acid.
D. Their pH varies upon standing for a long period of time.

37. The pH of a buffer solution prepared by using 0.04 M of C_6H_5COONa and 0.02 M C_6H_5COOH solutions is

(**Note**: K_a of acid is 6.4×10^{-5} at 25°C)

A. 3.89
B. 4.19
C. 4.49
D. 6.19

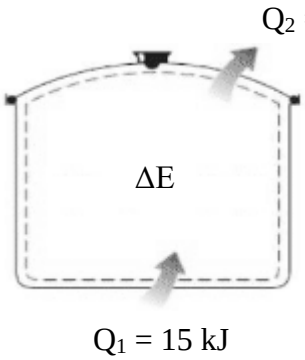
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38. The molarity of 0.25 dm^3 of a solution containing 10.35 g of K_2CO_3 is

(**Note**: Atomic mass of C = 12 amu, O = 16 amu, K = 39 amu)

A. 0.075 M.
B. 0.30 M.
C. 2.58 M.
D. 41.40 M.

39. Consider the given cooker as a system.



If no work is done by this system and the internal energy is zero, then the energy change (ΔE) of the system is

A. + 12 kJ.
B. – 12 kJ.
C. + 18 kJ.
D. – 18 kJ.

40. In a calorimeter, 13.3 moles of water are used to measure the heat of combustion from 0.303 g of sugar ($C_{12}H_{22}O_{11}$). The specific heat capacity of water is 4.2 $JK^{-1}g^{-1}$ and the rise in temperature is recorded as 5.0 K.

(**Note**: Molar masses of water and sugar are 18.0 g/ mol and 342.3 g/ mol respectively.)

The amount of heat evolved during combustion of 1.0 mol of sugar would be

A. 921.7 kJ/ mol.
B. 315.5 kJ/ mol.
C. 5027.4 kJ/ mol.
D. 5686.6 kJ/ mol.

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41. Given are the standard enthalpy changes for the series of reactions occurring in the formation of lithium fluoride (LiF).

Series of Reactions	Enthalpy Changes
$Li_{(s)} \rightarrow Li_{(g)}$	ΔH_{sub} = 155.2 kJ
$Li_{(g)} \rightarrow Li^{+}_{(g)} + e^{-}$	1st Ionization Energy of Li = 520 kJ
$\frac{1}{2}F_{2(g)} \rightarrow F_{(g)}$	$\frac{1}{2}$ Bond Energy of F_2 = 75.3 kJ
$F_{(g)} + e^{-} \rightarrow F^{-}_{(g)}$	Electron Affinity for Fluorine = −328 kJ
$Li^{+}_{(g)} + F^{-}_{(g)} \rightarrow LiF_{(s)}$	Lattice Energy of LiF = ?
$Li_{(s)} + \frac{1}{2}F_{2(g)} \rightarrow LiF_{(s)}$	Standard Enthalpy of Formation of $LiF_{(s)}$ = −594.1 kJ

The lattice energy of LiF is

A. –1016.6 kJ.
B. –1091.9 kJ.
C. –1610.7 kJ.
D. –1686.0 kJ.

42. Below are the thermochemical reactions for the oxidation of carbon and carbon monoxide with the associated enthalpy changes.

$C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)}$ $\Delta H = ?$

$C_{(s)} + \frac{1}{2}O_{2(g)} \longrightarrow CO_{(g)}$ $\Delta H_1 = -109\,kJmol^{-1}$

$CO_{(g)} + \frac{1}{2}O_{2(g)} \longrightarrow CO_{2(g)}$ $\Delta H_2 = -284.5\,kJmol^{-1}$

Based on the given information, the enthalpy of combustion for carbon forming carbon dioxide will be

A. + 393.5 $kJmol^{-1}$.
B. – 393.5 $kJmol^{-1}$.
C. + 175.5 $kJmol^{-1}$.
D. – 175.5 $kJmol^{-1}$.

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43. Given are the two half-cell reactions and their standard reduction potential.

$Cu^{2+}_{(aq)} + 2e^- \rightarrow Cu_{(s)}$ E^o = + 0.34 V

$Ni^{+2}_{(aq)} + 2e^- \rightarrow Ni_{(s)}$ E^o = − 0.25 V

The overall cell potential of the electrochemical cell will be

A. + 0.59 V.
B. − 0.59 V.
C. + 0.09 V.
D. − 0.09 V.

44. Which of the following statements is FALSE about a fuel cell?

A. It uses water as a source to produce electrical energy.
B. It generates large amount of energy to run a space vehicle.
C. It produces water that can be used for drinking by an astronaut.
D. It operates at high temperatures and does not produce pollutants.

45. An example of oxidation half reaction is

A. $Fe^{+2} \rightarrow Fe^{+3}$
B. $Cu^{+2} \rightarrow Cu^{+1}$
C. $Cl_2 \rightarrow 2Cl^{-1}$
D. $N_2 \rightarrow 2N^{-3}$

46. A current of 5 A is passed for 2 minutes during electrolysis.

The amount of charge transferred is

A. 10 C.
B. 120 C.
C. 300 C.
D. 600 C.

47. Which of the following is NOT a redox reaction?

A. $FeS_{(s)} + H_2SO_{4(aq)} \rightarrow FeSO_{4(aq)} + H_2S_{(g)}$
B. $H_2S_{(g)} + 4NaClO_{(aq)} \rightarrow H_2SO_{4(aq)} + 4NaCl_{(aq)}$
C. $Mg(OH)_{2(aq)} + H_2SO_{4(aq)} \rightarrow MgSO_{4(aq)} + 2H_2O_{(l)}$
D. $H_2O_{(l)} + SO_{2(g)} + NaOCl_{(aq)} \rightarrow H_2SO_{4(aq)} + NaCl_{(aq)}$

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48. Consider the given reaction.

$OF_{2(g)} + H_2O_{(l)} \rightarrow 2HF_{(g)} + O_{2(g)}$

The substance that acts as an oxidising agent in the given reaction is

A. O_2
B. HF
C. OF_2
D. H_2O

49. The oxidation number of iron in Fe_2O_3 is

A. + 1
B. + 2
C. + 3
D. + 6

50. The values of **v**, **w**, **x**, **y** and **z** that CORRECTLY balance the given chemical equation are

$vP_{(s)} + wHNO_{3(aq)} + xH_2O_{(l)} \rightarrow yH_3PO_{4(aq)} + zNO_{(g)}$

	v	**w**	**x**	**y**	**z**
A	2	1	1	5	3
B	5	3	3	7	5
C	3	7	5	1	5
D	3	5	2	3	5

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<u>ALTERNATE TO PRACTICAL</u> (ATP: Questions 51-65)

51. A student carries out a gravimetric analysis of 20 mL solution of $BaCl_2$ to estimate the amount of Ba^{2+} ions present in it. The molecular weight of the Ba^{2+} salt formed is 253.4 g and that of Ba^{2+} ions is 137.4 g.

The observation concluded during the experiment is as follows.

Weight of Filter Paper	**Weight of Filter Paper + Precipitates of Ba^{2+} Salt Formed**
0.9 g	1.8 g

With reference to the given observation, the amount of Ba^{2+} ions that will be present in 1000 mL of the solution is

A. 0.48 g.
B. 1.65 g.
C. 24.0 g.
D. 45.0 g.

52. After complete separation of a mixture through paper chromatography, its components can be identified by comparing the

A. colours of individual spots.
B. shapes of spots developed.
C. sizes of spots developed.
D. retention factor of spots.

53. The effect of common ion in a solution is that it

A. suppresses the dissociation of partially soluble salts.
B. decreases the number of ions in the solution.
C. shifts the equilibrium towards right.
D. increases the rate of reaction.

54. The addition of one compound to the solution of another compound having a common ion, causes those ions to precipitate that have

A. high molecular weight.
B. low solubility product.
C. high pH value.
D. low density.

55. A soap solution was prepared by using a sample of 8 g laundry soap. The calculated result of a titration between this soap solution and HCl shows that 1.4 g of free NaOH is overall utilised.

The percentage of free NaOH used in this titration is

A. 1.4%
B. 3.5%
C. 17.5%
D. 40.0%

56. If the value of **X** in $(COOH)_2.XH_2O$ is 2, then the amount of this compound in 1000 cm^3 of 1 M solution will be

(**Note**: $^{1}_{1}H$, $^{12}_{6}C$ and $^{16}_{8}O$)

A. 81.0 g.
B. 108.0 g.
C. 110.0 g.
D. 126.0 g.

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57. Uzma had 60 g impure sample of Na_2CO_3. She made a solution using 9 g of this sample in a 250 cm^3 volumetric flask. A titration was performed against the standard solution of HCl by taking 10 cm^3 of the solution formed and using methyl orange as an indicator. She determined that the solution contained 5.3 g of pure Na_2CO_3.

Based on the result, the amount of pure Na_2CO_3 in the entire sample will be

A. 51.0 g.
B. 47.1 g.
C. 35.3 g.
D. 9.6 g.

58. A student has to determine the molarity of H_2SO_4 through a volumetric analysis. Two different bases and two different indicators are mentioned in the table.

	Base	**Indicator**
I	NaOH	Methyl orange
II	Na_2CO_3	Methyl orange
III	NaOH	Phenolphthalein
IV	Na_2CO_3	Phenolphthalein

The CORRECT combination of base(s) and indicator(s) that he/ she could use for performing the experiment is/ are

A. I only.
B. I and II.
C. III only.
D. II and III.

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Use the given information to answer Q.59 and Q.60.

Zayan prepared a solution of oxalic acid at room temperature. He filtered and took 5 cm^3 of this solution to prepare 100 cm^3 solution in a volumetric flask. Finally, he pipetted out 10 cm^3 of 0.1 M standard NaOH solution and titrated it against 8 cm^3 of oxalic acid solution.

59. He calculated the concentration of oxalic acid in 8 cm^3 to be

A. 0.0625 M.
B. 0.08 M.
C. 0.04 M.
D. 0.125 M.

60. The solubility of the first solution that Zayan has prepared at room temperature is

(**Note**: $^{1}_{1}H$, $^{6}_{12}C$, $^{8}_{16}O$)

A. 5.04 g/ dm^3.
B. 10.08 g/ dm^3.
C. 12.6 g/ dm^3.
D. 15.75 g/ dm^3.

61. Consider the given acid and base pairs.

	Acid	Base
I	HCl	NaOH
II	HNO_3	NH_4OH
III	H_2CO_3	NH_4OH
IV	H_2SO_4	NaOH

The acid-base pairs which on neutralisation generate 13.7 kcal/mol energy are

A. I and II.
B. I and IV.
C. II and III.
D. III and IV.

62. During the standardisation of $KMnO_4$, the standard solution of oxalic acid with dilute H_2SO_4 is heated at 60°C because this temperature

A. increases the visibility of solution.
B. removes the impurity of oxalic acid.
C. speed up the production of Mn^{2+} ion.
D. requires high consumption of $KMnO_4$.

63. To determine the exact concentration of $KMnO_4$ solution it needs to be titrated against a standard solution because it

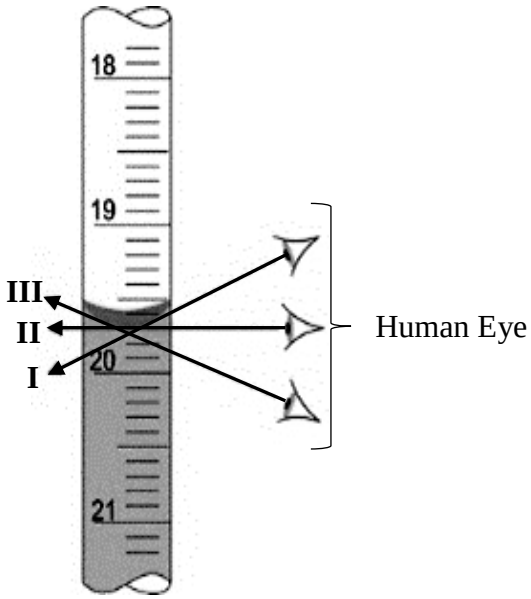
A. reacts in acidic medium.
B. releases nascent oxygen.
C. changes its concentration.
D. acts as an oxidising agent.

64. Which of the following statements is CORRECT about redox titration?

A. It involves proton transfer.
B. It typically requires an indicator.
C. It is carried out with transition elements.
D. It always produces water as a by-product.

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65. The given picture illustrates three ways of observing burette reading for a colourless solution.



The CORRECT way(s) to read a burette reading is/ are

A. I only.
B. II only.
C. I and III.
D. II and III.

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